

Appendix of Relevance-aware Individual Item Fairness Measures for Recommender Systems: Limitations and Usage Guidelines

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This is the appendix of the paper Relevance-aware Individual Item Fairness Measures for Recommender Systems: Limitations and Usage Guidelines (accepted to ACM TORS), which extends our SIGIR 2024 paper [1].

CCS Concepts: • **Information systems** → **Evaluation of retrieval results**; *Recommender systems*; • **General and reference** → *Evaluation*.

Additional Key Words and Phrases: relevance-aware fairness, item fairness, individual fairness, evaluation measures, recommender systems

A Mathematical workings for ranking strategies to achieve the (un)fairest JOINT scores

We provide the derivation of the fairest ranking for the original $IAA(u)$, $IFD_{\div}(u)$ (computed with $J_{\div-\text{our}}$), $IFD_{\times}(u)$, and $\Pi\text{-}F(u)$. For $IAA(u)$, we also show the workings behind the unfair ranking.

A.1 Inequity of Amortized Attention (IAA)

We show that ranking relevant items at the top- k (i.e., ranking items in non-increasing relevance) produces the fairest $\downarrow IAA(u)$. Let a_p^{IAA} be the exposure of item at position p , where item exposure in IAA is based on the product of the top- k indicator function and the examination function, i.e., $1_{L_u}(i) \cdot \tilde{e}_{li-\text{our}}(u, i)$; for brevity, we denote this as a_p . Let $r_{u,i} \in \{0, 1\}$ be the relevance of item i to user u ; for brevity, we denote this as r_i . Formally,

THEOREM A.1. *Given any two items i and i' , where $z(u, i) = p$, $z(u, i') = p'$, $r_i > r_{i'}$, ranking the item with higher relevance (i) closer to the top than the item with lower relevance (i') would result in a fairer (or equal) $\downarrow IAA(u)$ than doing vice versa:*

$$r_i > r_{i'} \wedge p < p' \Rightarrow |a_p - r_i| + |a_{p'} - r_{i'}| \leq |a_p - r_{i'}| + |a_{p'} - r_i|$$

PROOF. We prove Theorem A.1 by contradiction. Assume Theorem A.1 is false, i.e.:

$$\begin{aligned} & \neg (r_i > r_{i'} \wedge p < p' \Rightarrow |a_p - r_i| + |a_{p'} - r_{i'}| \leq |a_p - r_{i'}| + |a_{p'} - r_i|) \\ & \neg (\neg (r_i > r_{i'} \wedge p < p') \vee (|a_p - r_i| + |a_{p'} - r_{i'}| \leq |a_p - r_{i'}| + |a_{p'} - r_i|)) \\ & (r_i > r_{i'} \wedge p < p') \wedge (|a_p - r_i| + |a_{p'} - r_{i'}| > |a_p - r_{i'}| + |a_{p'} - r_i|) \end{aligned}$$

*Part of this work was done while visiting ADM+S at RMIT University.

As $r_i > r_{i'}$ and we deal with binary relevance, $r_i = 1$ and $r_{i'} = 0$. Substituting these values to the LHS of the inequality, we get: $|a_p - r_i| + |a_{p'} - r_{i'}| = |a_p - 1| + |a_{p'}| = |a_p - 1| + a_{p'}$ (as $a_{p'} \geq 0$). Substituting $r_i = 1$ and $r_{i'} = 0$ to the RHS of the inequality, we get: $|a_p - r_{i'}| + |a_{p'} - r_i| = |a_p| + |a_{p'} - 1| = a_p + |a_{p'} - 1|$ (as $a_p \geq 0$).

Putting the LHS and RHS of the inequality together, we have:

$$\begin{aligned}
& |a_p - 1| + a_{p'} > a_p + |a_{p'} - 1| \\
& |a_p - 1| - |a_{p'} - 1| > a_p - a_{p'} \\
& |a_p - 1| - |a_{p'} - 1| > 0 & (\because p < p' \Leftrightarrow a_p \geq a_{p'} \Leftrightarrow a_p - a_{p'} \geq 0) \\
& (1 - a_p) - (1 - a_{p'}) > 0 & (\because a_p \leq 1 \Leftrightarrow a_p - 1 \leq 0) \\
& a_{p'} > a_p & (\text{Contradiction with } p < p' \Leftrightarrow a_p \geq a_{p'})
\end{aligned}$$

Hence, Theorem A.1 holds. $\square$

Next, we show that ranking relevant items at the bottom (i.e., ranking items in non-decreasing relevance) produces the unfairer $\downarrow\text{IAA}(u)$.

THEOREM A.2. *Given any two items i and i' , where $z(u, i) = p$, $z(u, i') = p'$, $r_i < r_{i'}$, ranking the item with lower relevance (i) closer to the top than the item with higher relevance (i') would result in an unfairer (or equal) $\downarrow\text{IAA}(u)$ than doing vice versa:*

$$r_i < r_{i'} \wedge p < p' \Rightarrow |a_p - r_i| + |a_{p'} - r_{i'}| \geq |a_{p'} - r_i| + |a_p - r_{i'}|$$

PROOF. We prove Theorem A.2 by contradiction. Assume Theorem A.2 is false, i.e.:

$$\begin{aligned}
& \neg (r_i < r_{i'} \wedge p < p' \Rightarrow |a_p - r_i| + |a_{p'} - r_{i'}| \geq |a_{p'} - r_i| + |a_p - r_{i'}|) \\
& \neg (\neg (r_i < r_{i'} \wedge p < p') \vee (|a_p - r_i| + |a_{p'} - r_{i'}| \geq |a_{p'} - r_i| + |a_p - r_{i'}|)) \\
& (r_i < r_{i'} \wedge p < p') \wedge (|a_p - r_i| + |a_{p'} - r_{i'}| < |a_{p'} - r_i| + |a_p - r_{i'}|)
\end{aligned}$$

As $r_i < r_{i'}$ and we deal with binary relevance, $r_i = 0$ and $r_{i'} = 1$. Substituting these values to the left-hand side (LHS) of the inequality, we get: $|a_p - r_i| + |a_{p'} - r_{i'}| = |a_p| + |a_{p'} - 1| = a_p + |a_{p'} - 1|$ (as $a_p \geq 0$). Substituting $r_i = 0$ and $r_{i'} = 1$ to the right-hand side (RHS) of the inequality, we obtain: $|a_p - r_{i'}| + |a_{p'} - r_i| = |a_p - 1| + |a_{p'}| = |a_p - 1| + a_{p'}$ (as $a_{p'} \geq 0$). Putting the LHS and RHS of the inequality together, we have:

$$\begin{aligned}
& a_p + |a_{p'} - 1| < |a_p - 1| + a_{p'} \\
& a_p - a_{p'} < |a_p - 1| - |a_{p'} - 1| \\
& 0 < |a_p - 1| - |a_{p'} - 1| & (\because p < p' \Leftrightarrow a_p \geq a_{p'} \Leftrightarrow a_p - a_{p'} \geq 0) \\
& 0 < (1 - a_p) - (1 - a_{p'}) & (\because a_p \leq 1 \Leftrightarrow a_p - 1 \leq 0) \\
& a_p < a_{p'} & (\text{Contradiction with } p < p' \Leftrightarrow a_p \geq a_{p'})
\end{aligned}$$

Hence, Theorem A.2 holds. $\square$

A.2 Individual Fairness Disparity with Division (IFD_÷)

We show that ranking relevant items at the bottom (i.e., ranking items in non-decreasing relevance) produces the fairest $\downarrow\text{IFD}_{\div}(u)$. We denote as $a_p^{\text{IFD}_{\div}}$ the exposure of item at position p , where item exposure in IFD_÷ is based on $J_{\div\text{-ori}}$ (Eq. (27)). For brevity, we denote as a_p the item exposure.

THEOREM A.3. *Given any two relevant items i and i' , where $z(u, i) = p$, $z(u, i') = p'$, $r_i = r_{i'} = 1$, ranking the items closer to the bottom (beyond the top- k) would result in a fairer (or equal) $\downarrow\text{IFD}_{\div}(u)$ than doing vice versa. Formally,*

$$\max \left\{ 0, \frac{a_p}{r_i} - \frac{a_{p'}}{r_{i'}} \right\} + \max \left\{ 0, \frac{a_{p'}}{r_{i'}} - \frac{a_p}{r_i} \right\} \quad (\text{A1})$$

is: (1) lower for $k < p < p'$ than for $p < k < p'$ (ranking 1 item at the top- k and another one beyond the top- k); and (2) lower for $k < p < p'$ than for $p < p' < k$ (ranking both items at the top- k).

PROOF. We prove (1) and (2) separately. To prove (1), we start from the premise $k < p < p'$. Ranking the two relevant items beyond the top- k results in $a_p = a_{p'} = 0$. As such, Eq. (A1) is 0. Meanwhile, for $p < k < p'$, we have $a_p > 0$ and $a_{p'} = 0$. As $a_p > a_{p'}$, the expression $\frac{a_p}{r_i} - \frac{a_{p'}}{r_{i'}} = a_p > 0$, while $\frac{a_{p'}}{r_{i'}} - \frac{a_p}{r_i} = -a_p < 0$. Eq. (A1) evaluates to $\max\{0, a_p\} + \max\{0, -a_p\} = a_p > 0$. Therefore, (1) holds.

To prove (2), we use the results of proving (1), i.e., when $k < p < p'$, Eq. (A1) = 0. For $p < p' < k$, we have $a_p > a_{p'} > 0$. The expression $\frac{a_p}{r_i} - \frac{a_{p'}}{r_{i'}} = a_p - a_{p'} > 0$, while $\frac{a_{p'}}{r_{i'}} - \frac{a_p}{r_i} = a_{p'} - a_p < 0$. Eq. (A1) evaluates to $\max\{0, a_p - a_{p'}\} + \max\{0, a_{p'} - a_p\} = a_p - a_{p'} + 0 > 0$. Thus, (2) holds.

As both (1) and (2) hold, Theorem A.3 holds. $\square$

A.3 Individual Fairness Disparity with Multiplication (IFD_×)

We show that ranking relevant items at the bottom (i.e., ranking items in non-decreasing relevance) produces the fairest $\downarrow\text{IFD}_{\times}(u)$. We denote as $a_p^{\text{IFD}_{\times}}$ the exposure of item at position p , where item exposure in IFD_× is based on $J_{\times\text{-ori}}$ (Eq. (4)). For brevity, we denote as a_p the item exposure.

THEOREM A.4. *Given any two items i and i' , where $z(u, i) = p$, $z(u, i') = p'$, $r_i < r_{i'}$, ranking the item with lower relevance (i) closer to the top than the item with higher relevance (i') would result in a fairer (or equal) $\downarrow\text{IFD}_{\times}(u)$ than doing vice versa:*

$$r_i < r_{i'} \wedge p < p' \Rightarrow (a_p r_i - a_{p'} r_{i'})^2 \leq (a_{p'} r_i - a_p r_{i'})^2$$

PROOF. We prove Theorem A.4 by contradiction. Assume Theorem A.4 is false, i.e.:

$$\begin{aligned} & \neg \left(r_i < r_{i'} \wedge p < p' \Rightarrow (a_p r_i - a_{p'} r_{i'})^2 \leq (a_{p'} r_i - a_p r_{i'})^2 \right) \\ & \neg \left(\neg (r_i < r_{i'} \wedge p < p') \vee [(a_p r_i - a_{p'} r_{i'})^2 \leq (a_{p'} r_i - a_p r_{i'})^2] \right) \\ & (r_i < r_{i'} \wedge p < p') \wedge [(a_p r_i - a_{p'} r_{i'})^2 > (a_{p'} r_i - a_p r_{i'})^2] \end{aligned}$$

As $r_i < r_{i'}$ and we deal with binary relevance, $r_i = 0$ and $r_{i'} = 1$. Substituting these values to the inequality, we get $a_{p'}^2 > a_p^2$. There are three possibilities of how the rank positions p, p' relate to the cut-off k :

- (1) Both rank positions are beyond k , $k < p < p'$. In this case, $a_p = a_{p'} = 0$ as items at position p and p' are unexposed. As $a_{p'}^2 = a_p^2$, this contradicts $a_{p'}^2 > a_p^2$.
- (2) Both rank positions are at the top- k , $p < p' < k$. Here, $a_p > a_{p'} > 0 \Leftrightarrow a_p^2 > a_{p'}^2$, which contradicts $a_{p'}^2 > a_p^2$.

- (3) One rank position is at the top- k , and another one beyond top- k , $p < k < p'$. Here, $a_p > 0 = a_{p'} \Leftrightarrow a_p^2 > a_{p'}^2$, which contradicts $a_{p'}^2 > a_p^2$.

Therefore, Theorem A.4 holds via proof by contradiction. $\square$

A.4 Individual-user-to-Individual-item Fairness (II-F)

We show that ranking relevant items at the top- k (i.e., ranking items in non-increasing relevance) produces the fairest $\downarrow$ II-F(u). We denote as $a_p^{\text{II-F}}$ the exposure of item at position p , where item exposure in II-F is $E_{u,i}$ (Eq. (22)). For brevity, we denote as a_p the item exposure and as t_i the target exposure of item i , $E_{u,i}^*$ (Eq. (23)).

THEOREM A.5. *Given any two items i and i' , where $z(u, i) = p$, $z(u, i') = p'$, $r_i > r_{i'}$, ranking the item with higher relevance (i) closer to the top than the item with lower relevance (i') would result in a fairer (or equal) $\downarrow$ II-F(u) than doing vice versa:*

$$r_i > r_{i'} \wedge p < p' \Rightarrow (a_p - t_i)^2 + (a_{p'} - t_{i'})^2 \leq (a_{p'} - t_i)^2 + (a_p - t_{i'})^2$$

PROOF. We prove Theorem A.5 by contradiction. Assume Theorem A.5 is false, i.e.:

$$\begin{aligned} & \neg \left(r_i > r_{i'} \wedge p < p' \Rightarrow (a_p - t_i)^2 + (a_{p'} - t_{i'})^2 \leq (a_{p'} - t_i)^2 + (a_p - t_{i'})^2 \right) \\ & \neg \left(\neg (r_i > r_{i'} \wedge p < p') \vee \left[(a_p - t_i)^2 + (a_{p'} - t_{i'})^2 \leq (a_{p'} - t_i)^2 + (a_p - t_{i'})^2 \right] \right) \\ & (r_i > r_{i'} \wedge p < p') \wedge \left[(a_p - t_i)^2 + (a_{p'} - t_{i'})^2 > (a_{p'} - t_i)^2 + (a_p - t_{i'})^2 \right] \end{aligned}$$

As $r_i > r_{i'}$ and we deal with binary relevance, $r_i = 1$ and $r_{i'} = 0$, which means the target exposures of item i and i' are $t_i > t_{i'} = 0$. Substituting these values to the inequality, we get:

$$\begin{aligned} & (a_p - t_i)^2 + a_{p'}^2 > (a_{p'} - t_i)^2 + a_p^2 \\ & (a_p - t_i)^2 - (a_{p'} - t_i)^2 > a_p^2 - a_{p'}^2 \\ & (a_p^2 - 2a_p t_i + t_i^2) - (a_{p'}^2 - 2a_{p'} t_i + t_i^2) > a_p^2 - a_{p'}^2 \\ & (-2a_p t_i) - (-2a_{p'} t_i) > 0 \\ & 2a_{p'} t_i > 2a_p t_i \\ & a_{p'} > a_p \end{aligned}$$

However, as $p < p' \Leftrightarrow a_p \geq a_{p'}$, there is a contradiction. Hence, Theorem A.5 holds. $\square$

References

- [1] Theresia Veronika Rampisela, Tuukka Ruotsalo, Maria Maistro, and Christina Lioma. 2024. Can We Trust Recommender System Fairness Evaluation? The Role of Fairness and Relevance. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '24)*. Association for Computing Machinery, New York, NY, USA, 271–281. doi:10.1145/3626772.3657832